E-commerce Sales

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
data=pd.read_csv(r"C:\Users\user\Desktop\Downloads\Ecommerce\Sample - Superstore.csv",encoding='latin-1')
```

```
data.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.9600
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.6200
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.3680

5 rows × 21 columns

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

```
data.describe()
```

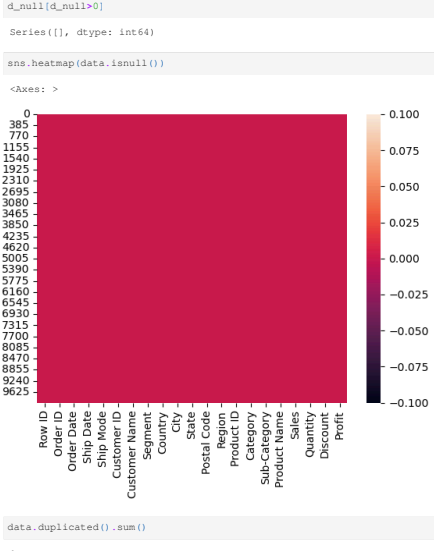
	Row ID	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000

```
d_null=data.isnull().sum()
d_null[d_null>0]
```

```
Series([], dtype: int64)
```

```
sns.heatmap(data.isnull())
```

```
<Axes: >
```



```
data.duplicated().sum()
```

```
0
```

```
data_object = data[[ 'Ship Mode', 'Customer ID', 'Customer Name',
           'Segment', 'Country', 'City', 'State', 'Region',
           'Product ID', 'Category', 'Sub-Category', 'Product Name']]
data_object.head(20)
```

	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Region	Product ID	Category	Sub-Category	Product Name
0	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase
1	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...
2	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...
3	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table
4	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System
5	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Access...
6	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	West	OFF-AR-10002833	Office Supplies	Art	Newell 322
7	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone
8	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by S...
9	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge
10	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables
11	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	West	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black
12	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967
13	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine
14	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	Central	OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner,...
15	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	Central	OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders
16	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	Central	OFF-ST-10004186	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x...
17	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer
18	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	West	OFF-AR-10003056	Office Supplies	Art	Newell 341
19	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone

```
for feature in data_object:
    print(feature,':',data_object[feature].nunique())
```

```
Ship Mode : 4
Customer ID : 793
Customer Name : 793
Segment : 3
Country : 1
City : 531
State : 49
Region : 4
Product ID : 1862
Category : 3
Sub-Category : 17
Product Name : 1850
```

```
data['Order Date'].nunique()
```

```
1237
```

```
data['Ship Date'].nunique()
```

```
1334
```

EDA

```
data.columns=data.columns.str.lower()
data.head(10)
```

	row id	order id	order date	ship date	ship mode	customer id	customer name	segment	country	city	...	postal code	region	product id	category	sub-category	product name	sales
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.9600
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.6200
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.3680
5	6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	...	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Access...	48.8600
6	7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	...	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.2800
7	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	...	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.1520
8	9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	...	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by S...	18.5040
9	10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	...	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9000

10 rows × 21 columns

```
#Top Selling products
data["product name"].value_counts()
```

```
product name
Staple envelope                                                                48
Staples                                                                        46
Easy-staple paper                                                              46
Avery Non-Stick Binders                                                        20
Staples in misc. colors                                                        19
                                                                               ..
Boston 1900 Electric Pencil Sharpener                                           1
RCA ViSYS 25423RE1 Corded phone                                                 1
Canon Color ImageCLASS MF8580Cdw Wireless Laser All-In-One Printer, Copier, Scanner   1
Newell 342                                                                      1
Eldon Jumbo ProFile Portable File Boxes Graphite/Black                          1
Name: count, Length: 1850, dtype: int64
```

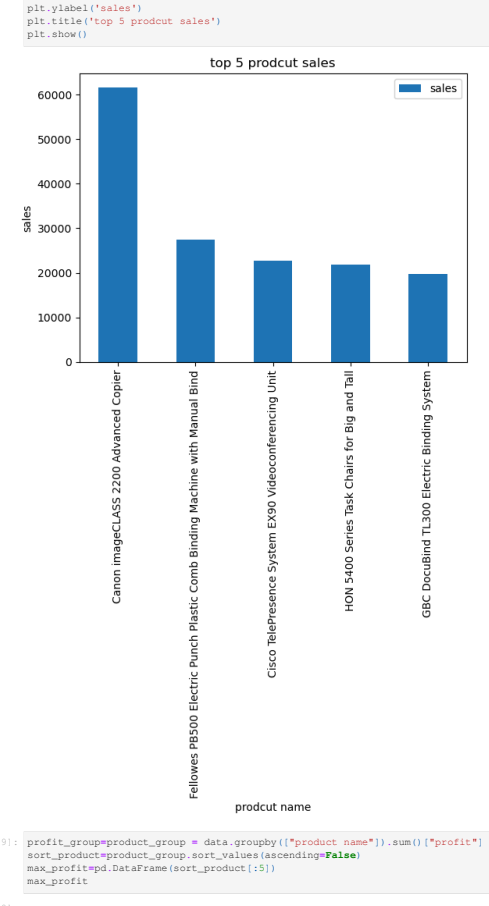
```
product_group = data.groupby(["product name"]).sum()["sales"]
sort_product=product_group.sort_values(ascending=False)
sort_product
```

```
product name
Canon imageCLASS 2200 Advanced Copier                                      61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind 27453.384
Cisco TelePresence System EX90 Videoconferencing Unit                      22638.480
HON 5400 Series Task Chairs for Big and Tall                               21870.576
GBC DocuBind TL300 Electric Binding System                                 19823.479
                                                                                 ...
Avery Hi-Liter Pen Style Six-Color Fluorescent Set                             7.700
Grip Seal Envelopes                                                            7.072
Xerox 20                                                                       6.480
Avery 5                                                                        5.760
Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac               1.624
Name: sales, Length: 1850, dtype: float64
```

```
max_product=pd.DataFrame(sort_product[:5])
max_product
```

product name	sales
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.384
Cisco TelePresence System EX90 Videoconferencing Unit	22638.480
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479

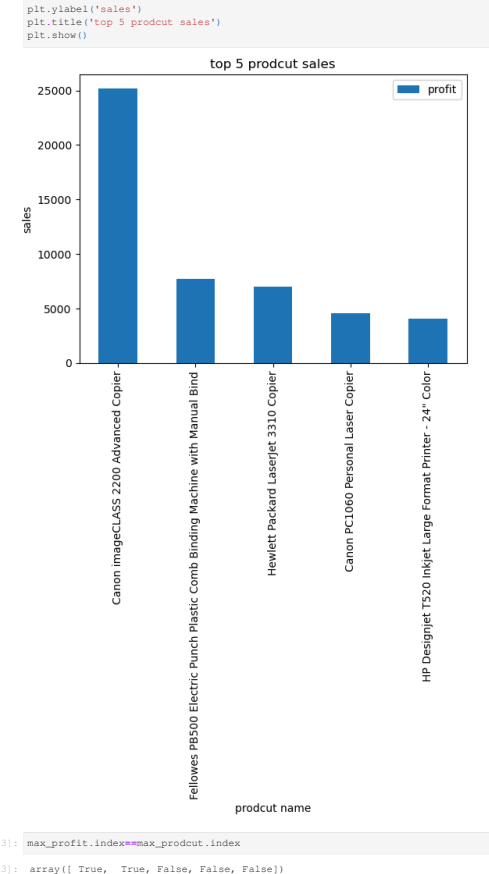
```
max_product.plot(kind="bar")
plt.xlabel("prodcut name")
plt.ylabel("sales")
plt.title("top 5 prodcut sales")
plt.show()
```



```
profit_group=product_group = data.groupby(["product name"]).sum()["profit"]
sort_product=product_group.sort_values(ascending=False)
max_profit=pd.DataFrame(sort_product[:5])
max_profit
```

product name	profit
Canon imageCLASS 2200 Advanced Copier	25199.9280
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	7753.0390
Hewlett Packard LaserJet 3310 Copier	6983.8836
Canon PC1060 Personal Laser Copier	4570.9347
HP Designjet T520 Inkjet Large Format Printer - 24" Color	4094.9766

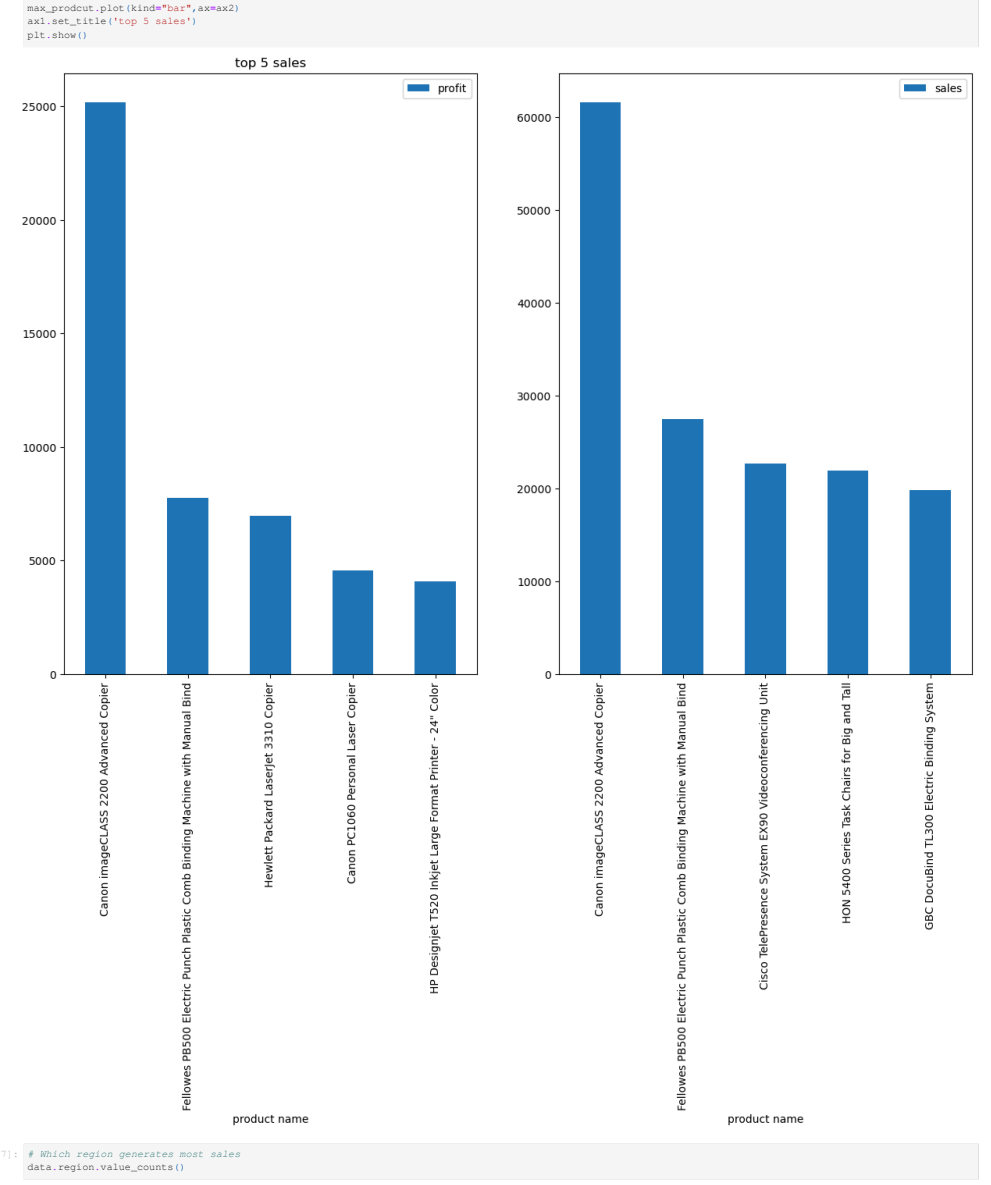
```
max_profit.plot(kind="bar")
plt.xlabel("prodcut name")
plt.ylabel("sales")
plt.title("top 5 prodcut sales")
plt.show()
```



```
max_profit.index==max_product.index
```

```
array([ True,  True, False, False, False])
```

```
fig,(ax1,ax2)=plt.subplots(1,2,figsize=(15,10))
max_profit.plot(kind="bar",ax=ax1)
ax1.set_title("top 5 profit")
max_product.plot(kind="bar",ax=ax2)
ax1.set_title("top 5 sales")
plt.show()
```



```
# Which region generates most sales
data.region.value_counts()
```

```
region
West       3203
East       2848
Central    2323
South      1620
Name: count, dtype: int64
```

```
data.groupby(data.region).sum()["profit"]
```

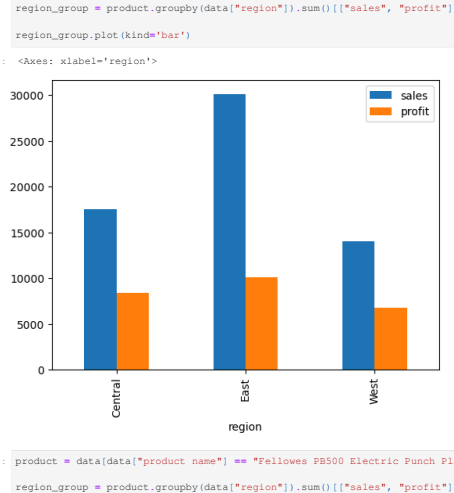
```
region
Central     39706.3625
East        91522.7800
South       46749.4303
West       108418.4489
Name: profit, dtype: float64
```

```
data.region.groupby(data["product name"]).value_counts()
```

```
product name                                         region
"While you Were Out" Message Book, One Form per Page  Central    1
                                                      South      1
                                                      West       1
#10 Gummed Flap White Envelopes, 100/Box              Central    2
                                                      East       2
                                                                ..
invisibleSHIELD by ZAGG Smudge-Free Screen Protector  West       3
                                                      Central    1
netTALK DUO VoIP Telephone Service                    West       5
                                                      Central    1
                                                      East       1
Name: count, Length: 5255, dtype: int64
```

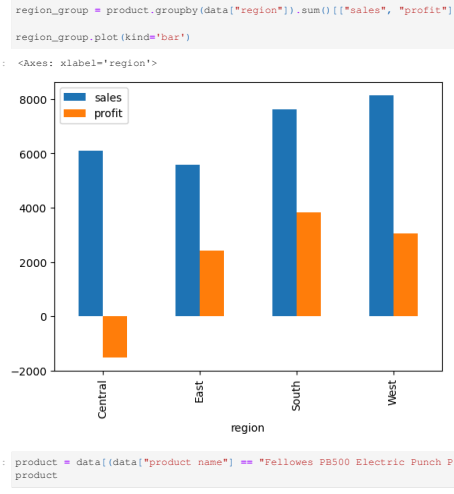
```
product = data[data["product name"] == "Canon imageCLASS 2200 Advanced Copier"]
region_group = product.groupby(data["region"]).sum()[["sales", "profit"]]

region_group.plot(kind='bar')
```

```
<Axes: xlabel='region'>
```



```
product = data[data["product name"] == "Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind"]

region_group = product.groupby(data["region"]).sum()[["sales", "profit"]]

region_group.plot(kind="bar")
```

```
<Axes: xlabel='region'>
```

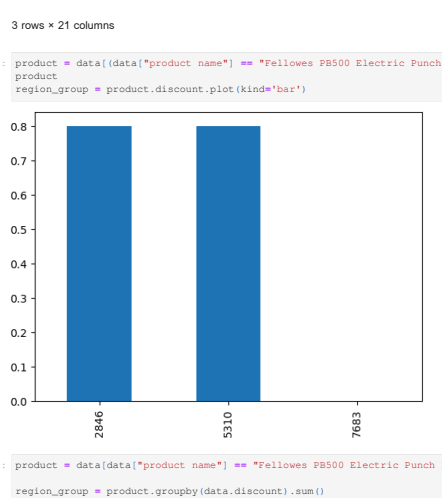


```
product = data[(data["product name"] == "Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind" ) &(data.region=="Central")]
product
```

	row id	order id	order date	ship date	ship mode	customer id	customer name	segment	country	city	...	postal code	region	product id	category	sub-category	product name	sales	quantity
2846	2847	CA-2017-152093	9/10/2017	9/15/2017	Standard Class	SN-20560	Skye Norling	Home Office	United States	Chicago	...	60653	Central	OFF-BI-10003527	Office Supplies	Binders	Fellowes PB500 Electric Punch Plastic Comb Bin...	762.594	3
5310	5311	CA-2017-131254	11/19/2017	11/21/2017	First Class	NC-18415	Nathan Cano	Consumer	United States	Houston	...	77095	Central	OFF-BI-10003527	Office Supplies	Binders	Fellowes PB500 Electric Punch Plastic Comb Bin...	1525.188	6
7683	7684	CA-2015-120782	4/28/2015	5/1/2015	First Class	SD-20485	Shirley Daniels	Home Office	United States	Midland	...	48640	Central	OFF-BI-10003527	Office Supplies	Binders	Fellowes PB500 Electric Punch Plastic Comb Bin...	3812.970	3

3 rows × 21 columns

```
product = data[(data["product name"] == "Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind" ) &(data.region=="Central")]
product
region_group = product.discount.plot(kind="bar")
```



```
product = data[data["product name"] == "Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind"]

region_group = product.groupby(data.discount).sum()
region_group
```

discount	row id	order id	order date	ship date	ship mode	customer id	customer name
0.0	24236	CA-2015-145352CA-2015-162782CA-2015-120782CA-2...	3/16/20152/21/20154/28/20151/31/2016	3/22/20152/27/20155/1/20152/6/2016	Standard ClassStandard ClassFirst ClassStandar...	CM-12385PW-19240SD-20485BF-11170	Christopher MartinezPierre WenerShirley Daniel...
0.2	29465	CA-2016-146633US-2015-163825CA-2017-135909CA-2...	11/15/20166/16/201510/13/201711/11/2014	11/17/20166/19/201510/20/201711/13/2014	Second ClassFirst ClassStandard ClassFirst Class	TG-21310LC-16885JW-15220MZ-17515	Toby GradeLena CreightonJane WacoMary Zewe
0.8	8158	CA-2017-152093CA-2017-131254	9/10/201711/19/2017	9/15/201711/21/2017	Standard ClassFirst Class	SN-20560NC-18415	Skye NorlingNathan Cano

3 rows × 21 columns

```
# Sales Trend Over Time Monthly & Yearly
monthly_sales=data.groupby(["order date"],as_index=False).sum()
monthly_sales
```

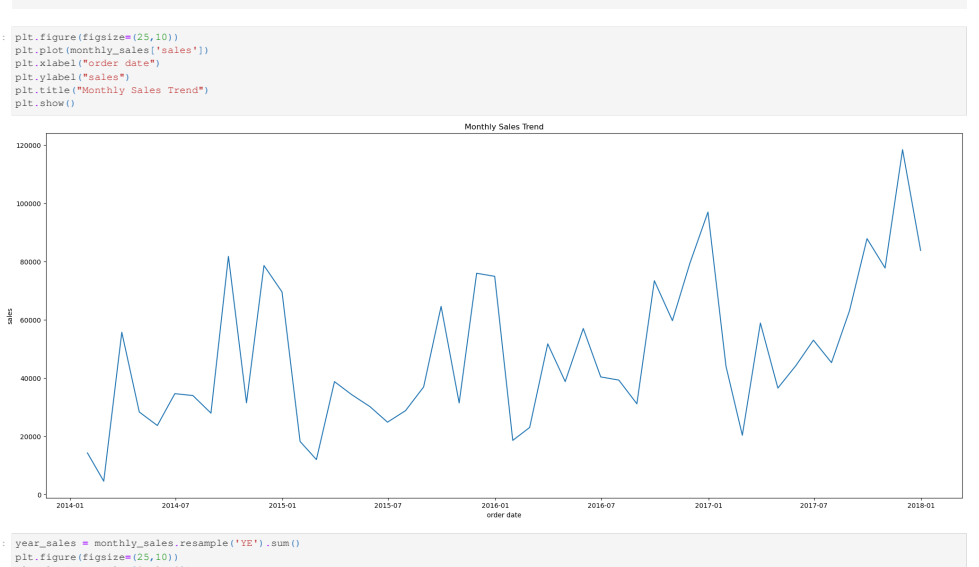
	order date	row id	order id	ship date	ship mode	customer id	customer name	segment
0	1/1/2017	68053	CA-2017-107503CA-2017-144463CA-2017-154466CA-2...	1/6/20171/5/20171/2/20171/5/20171/5/20...	Standard ClassStandard ClassFirst ClassStandar...	GA-14725SC-20725DP-13390JM-15250JM-152...	Guy ArmstrongSteven CartwrightDennis PardueJan...	ConsumerConsumerHome OfficeConsumerCon...
1	1/10/2014	1733	CA-2014-149020CA-2014-149020	1/15/20141/15/2014	Standard ClassStandard Class	AJ-10780AJ-10780	Anthony JacobsAnthony Jacobs	CorporateCorporate
2	1/10/2015	1670	US-2015-157154	1/15/2015	Standard Class	MM-17920	Michael Moore	Consumer
3	1/10/2016	16305	CA-2016-116547CA-2016-116547CA-2016-163937	1/17/20161/17/20161/13/2016	Standard ClassStandard ClassSecond Class	KB-16585KB-16585JB-16000	Ken BlackKen BlackJoy Bell-	CorporateCorporateConsumer
4	1/11/2014	717	CA-2014-130092	1/14/2014	First Class	SV-20365	Seth Vernon	Consumer
...	...	...	...	...	...	...	...	...
1232	9/8/2016	81920	CA-2016-108987CA-2016-108987CA-2016-108987CA-2...	9/10/20169/10/20169/10/20169/10/20169...	Second ClassSecond ClassSecond ClassSecond Cla...	AG-10675AG-10675AG-10675AG-10675VM-216...	Anna GaymanAnna GaymanAnna GaymanAnna GaymanVa...	ConsumerConsumerConsumerConsumerHome OfficeHom...
1233	9/8/2017	83975	US-2017-124968CA-2017-149895CA-2017-158036CA-2...	9/13/20179/12/20179/14/20179/10/20179/12/20179...	Second ClassStandard ClassStandard ClassSecond...	MM-18055EB-14110DB-13555RS-19765AR-10345BF-110...	Michelle MoorayEugene BarchasDorothy BaddersRol...	ConsumerConsumerCorporateCorporateCorporateCor...
1234	9/9/2014	58045	CA-2014-144407CA-2014-126403CA-2014-126403CA-2...	9/15/20149/12/20149/12/20149/12/20149...	Standard ClassSecond ClassSecond ClassSecond C...	MS-17365RR-19525RR-19525RR-19525LC-171...	Maribeth SchnellingRick ReedRick ReedRick Reed...	ConsumerCorporateCorporateCorporateCo...
1235	9/9/2016	7555	CA-2016-110023CA-2016-112893CA-2016-137176	9/11/20169/13/20169/14/2016	First ClassSecond ClassStandard Class	TS-21610AT-10735DB-12910	Troy StaebelAnnie ThurmanDaniel Byrd	ConsumerConsumerHome Office
1236	9/9/2017	136887	US-2017-118087US-2017-118087US-2017-118087US-2...	9/13/20179/13/20179/13/20179/13/20179...	Standard ClassStandard ClassStandard ClassStan...	SP-20620SP-20620SP-20620SP-20620SP-206...	Stefania PerrinoStefania PerrinoStefania Perri...	CorporateCorporateCorporateCorporateC...

1237 rows × 21 columns

```
monthly_sales = monthly_sales.set_index('order date')
monthly_sales.index = pd.to_datetime(monthly_sales.index)
monthly_sales = monthly_sales.resample('ME').sum()
monthly_sales
```

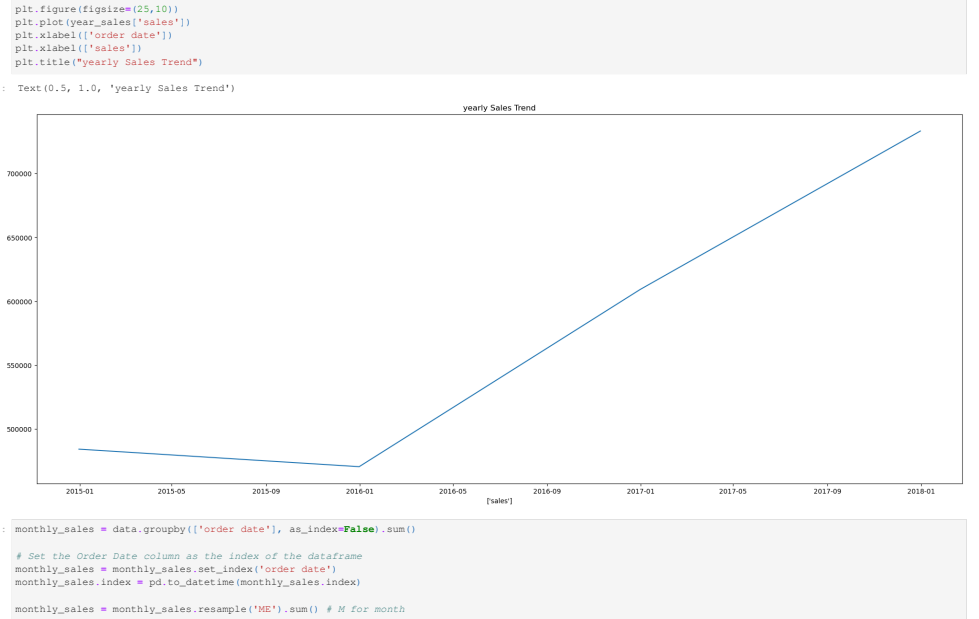
order date	row id	order id	ship date	ship mode	customer id	customer name	segment
2014-01-31	346176	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-02-28	272235	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-03-31	671575	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-04-30	690625	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-05-31	611708	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-06-30	491794	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-07-31	746403	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-08-31	729301	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-09-30	1298740	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-10-31	865655	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-11-30	1605852	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2014-12-31	1373951	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-01-31	290696	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-02-28	307146	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-03-31	695521	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-04-30	773461	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-05-31	613409	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-06-30	769300	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-07-31	584608	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-08-31	826534	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-09-30	1451814	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-10-31	782639	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-11-30	1639831	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2015-12-31	1478539	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-01-31	477972	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-02-29	428663	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-03-31	615506	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-04-30	706340	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-05-31	1101329	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-06-30	923881	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-07-31	980979	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-08-31	871993	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-09-30	1690365	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-10-31	982967	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-11-30	1760607	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2016-12-31	1818182	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-01-31	847373	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-02-28	593504	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-03-31	1293993	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-04-30	948011	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-05-31	1346912	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-06-30	1171681	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-07-31	1183701	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-08-31	1141105	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-09-30	2358436	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-10-31	1412854	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-11-30	2300116	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2017-12-31	2250814	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

```
plt.figure(figsize=(25,10))
plt.plot(monthly_sales["sales"])
plt.xlabel("order date")
plt.ylabel("sales")
plt.title("Monthly Sales Trend")
plt.show()
```

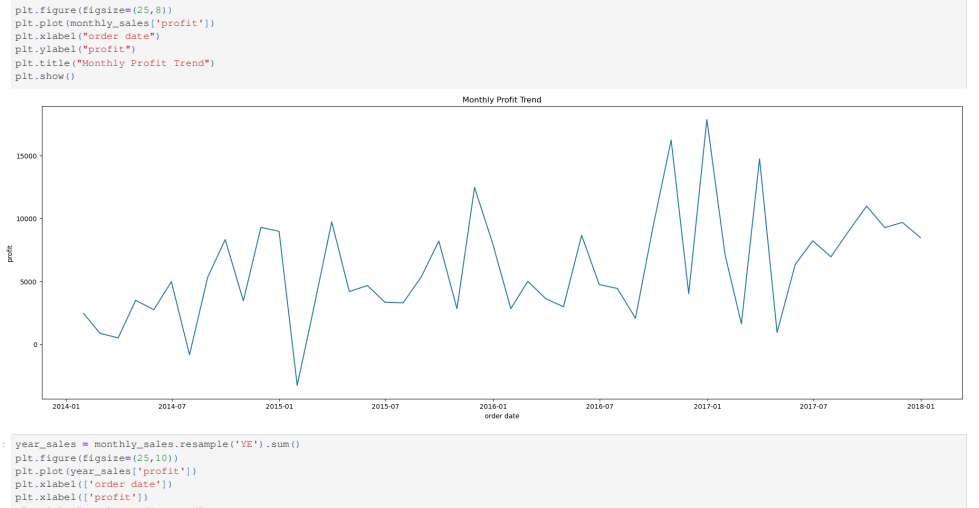


```
year_sales = monthly_sales.resample("YE").sum()
plt.figure(figsize=(25,10))
plt.plot(year_sales["sales"])
plt.xlabel(["order date"])
plt.ylabel(["sales"])
plt.title("yearly Sales Trend")
```

```
Text(0.5, 1.0, 'yearly Sales Trend')
```

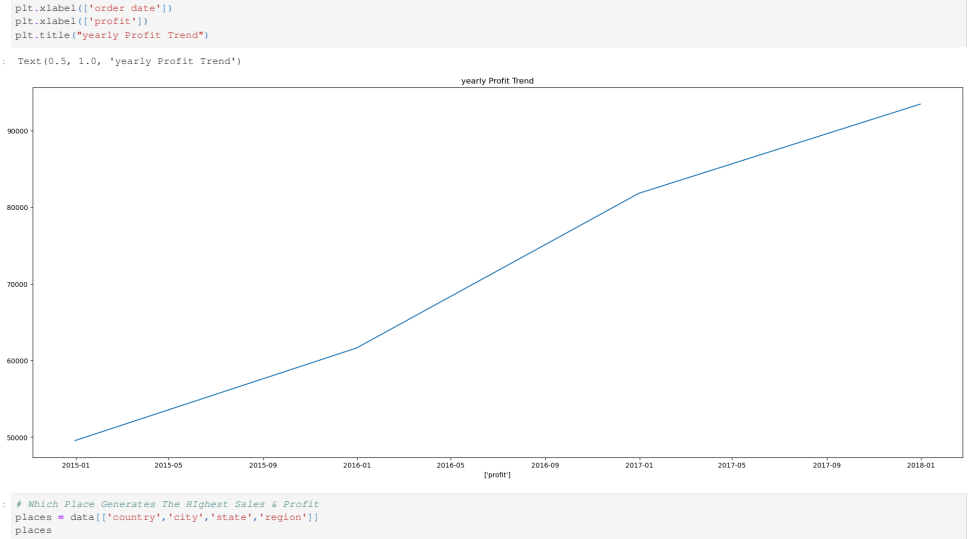


```
monthly_sales = data.groupby(["order date"], as_index=False).sum()

# Set the Order Date column as the index of the dataframe
monthly_sales = monthly_sales.set_index("order date")
monthly_sales.index = pd.to_datetime(monthly_sales.index)

monthly_sales = monthly_sales.resample("ME").sum() # # For month

# Plot
plt.figure(figsize=(25,8))
plt.plot(monthly_sales["profit"])
plt.xlabel("order date")
plt.ylabel("profit")
plt.title("Monthly Profit Trend")
plt.show()
```



```
year_sales = monthly_sales.resample("YE").sum()
plt.figure(figsize=(25,10))
plt.plot(year_sales["profit"])
plt.xlabel(["order date"])
plt.ylabel(["profit"])
plt.title("yearly Profit Trend")
```

```
Text(0.5, 1.0, 'yearly Profit Trend')
```



```
# Which Place Generates The Highest Sales & Profit
places = data[["country","city","state","region"]]
places
```

	country	city	state	region
0	United States	Henderson	Kentucky	South
1	United States	Henderson	Kentucky	South
2	United States	Los Angeles	California	West
3	United States	Fort Lauderdale	Florida	South
4	United States	Fort Lauderdale	Florida	South
...	...	...	...	...
9989	United States	Miami	Florida	South
9990	United States	Costa Mesa	California	West
9991	United States	Costa Mesa	California	West
9992	United States	Costa Mesa	California	West
9993	United States	Westminster	California	West

9994 rows × 4 columns

```
for place in places.columns:
    print(place,":",places[place].nunique())
```

```
country : 1
city : 531
state : 49
region : 4
```

```
places = data[["country","city","state","region","sales","profit"]]
```

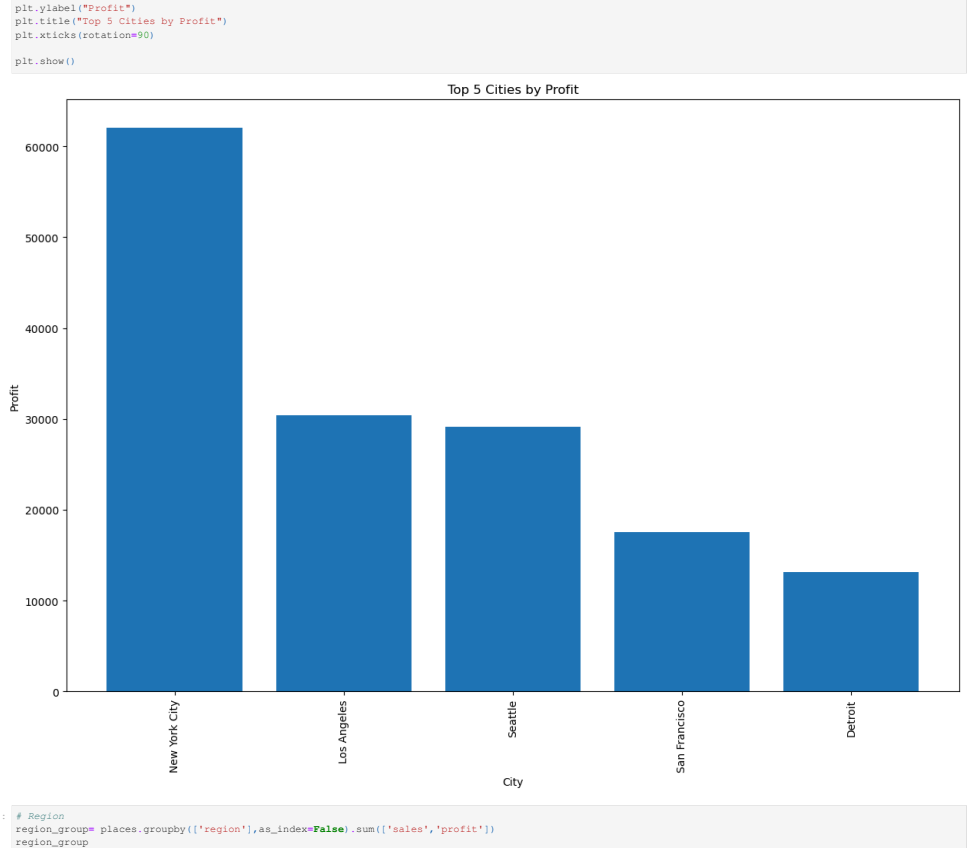
```
# Country
places_group= places.groupby(["country"]).sum()[["sales","profit"]]
places_group
```

country	sales	profit
United States	2.297201e+06	286397.0217

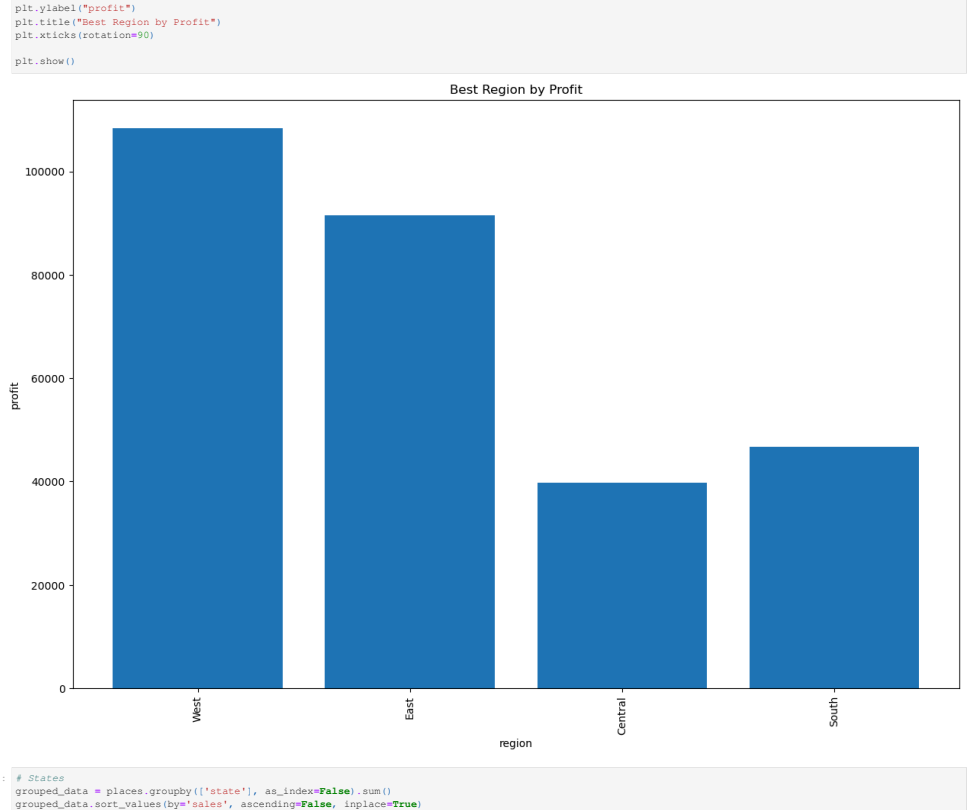
```
# City
city_group= places.groupby(["city"],as_index=False).sum()[["sales","profit"]]
city_group
city_group.sort_values(by="profit", ascending=False, inplace=True)

top_city=pd.DataFrame(city_group[:5])
top_city
```

	city	sales	profit
329	New York City	256368.161	62036.9837
266	Los Angeles	175851.341	30440.7579
452	Seattle	119540.742	29156.0967
438	San Francisco	112669.092	17507.3854
123	Detroit	42446.944	13181.7908

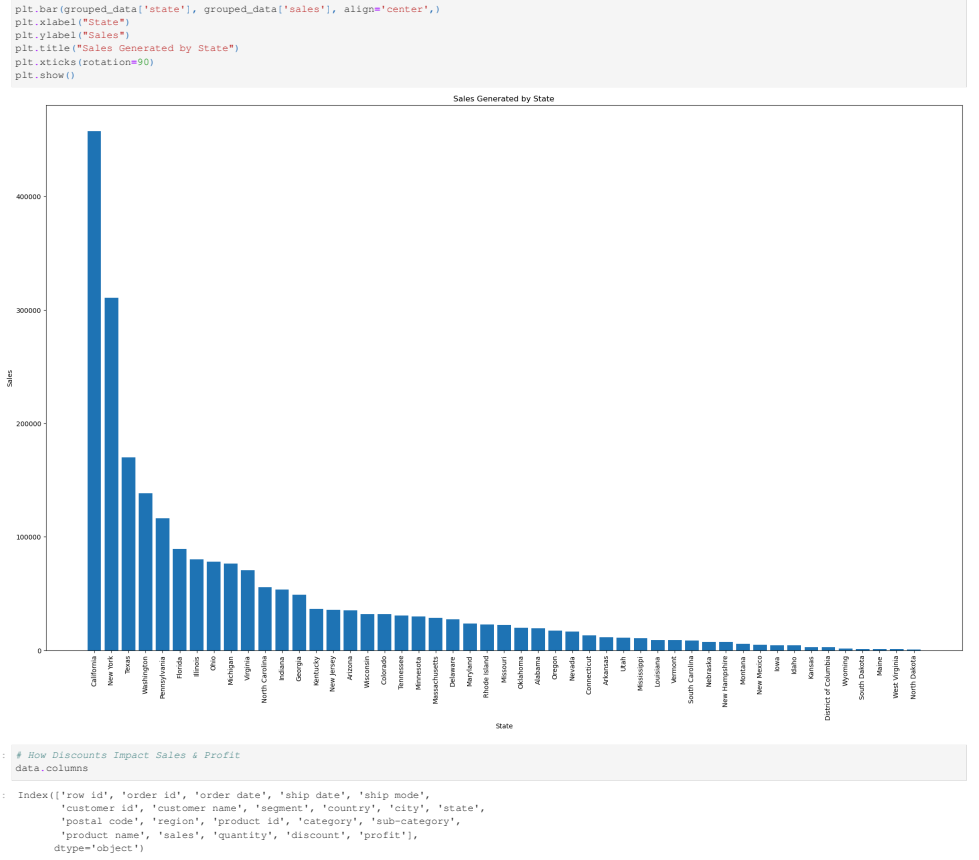
```
plt.figure(figsize=(15,10))
plt.bar(top_city["city"], top_city["profit"], align="center")
plt.xlabel("City")
plt.ylabel("Profit")
plt.title("Top 5 Cities by Profit")
plt.xticks(rotation=90)

plt.show()
```



```
# Region
region_group= places.groupby(["region"],as_index=False).sum()[["sales","profit"]]
region_group

region_group.sort_values(by="sales", ascending=False, inplace=True)
top_region=pd.DataFrame(region_group[:5])
top_region

plt.figure(figsize=(15,10))
plt.bar(top_region["region"], top_region["profit"], align="center")
plt.xlabel("region")
plt.ylabel("profit")
plt.title("Best Region by Profit")
plt.xticks(rotation=90)

plt.show()
```



```
# States
grouped_data = places.groupby(["state"], as_index=False).sum()
grouped_data.sort_values(by="sales", ascending=False, inplace=True)
plt.figure(figsize=(25,15))
plt.bar(grouped_data["state"], grouped_data["sales"], align="center",)
plt.xlabel("State")
plt.ylabel("Sales")
plt.title("Sales Generated by State")
plt.xticks(rotation=90)

plt.show()
```



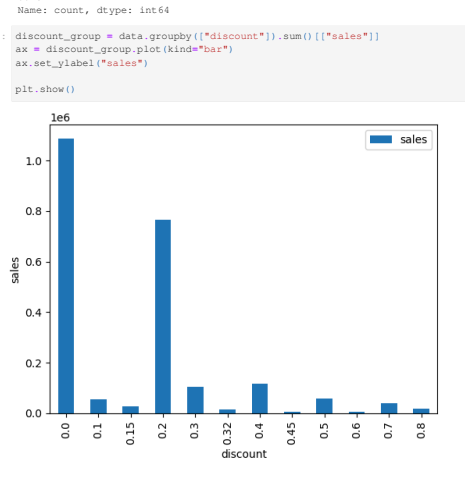
```
# How Discounts Impact Sales & Profit
data.columns
```

```
Index(['row id', 'order id', 'order date', 'ship date', 'ship mode',
       'customer id', 'customer name', 'segment', 'country', 'city', 'state',
       'postal code', 'region', 'product id', 'category', 'sub-category',
       'product name', 'sales', 'quantity', 'discount', 'profit'],
      dtype='object')
```

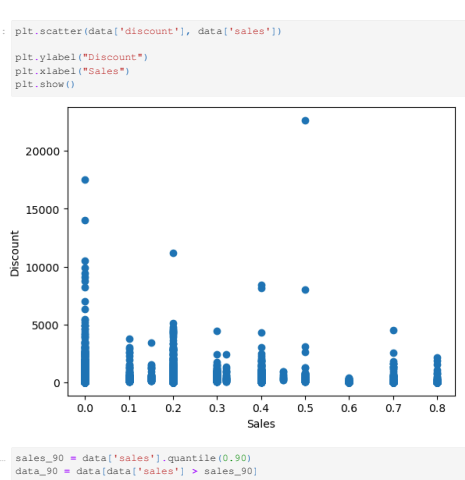
```
data.discount.value_counts()
```

```
discount
0.00    4798
0.20    3657
0.70     418
0.80     300
0.30     227
0.40     206
0.60     138
0.10      94
0.50      66
0.15      52
0.32      27
0.45      11
Name: count, dtype: int64
```

```
discount_group = data.groupby(["discount"]).sum()[["sales"]]
ax = discount_group.plot(kind="bar")
ax.set_ylabel("sales")

plt.show()
```



```
plt.scatter(data["discount"], data["sales"])

plt.ylabel("Discount")
plt.xlabel("Sales")
plt.show()
```



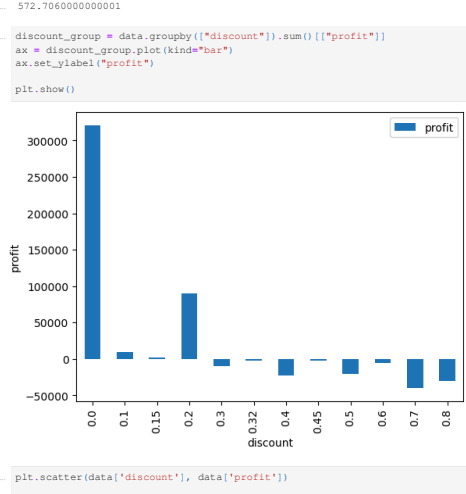
```
sales_90 = data["sales"].quantile(0.90)
data_90 = data[data["sales"] > sales_90]
data_90.shape
```

```
(1000, 21)
```

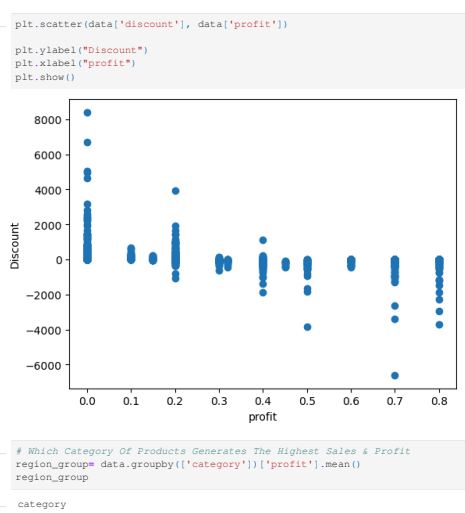
```
sales_90
```

```
572.7060000000001
```

```
discount_group = data.groupby(["discount"]).sum()[["profit"]]
ax = discount_group.plot(kind="bar")
ax.set_ylabel("profit")

plt.show()
```



```
plt.scatter(data["discount"], data["profit"])

plt.ylabel("Discount")
plt.xlabel("profit")
plt.show()
```



```
# Which Category Of Products Generates The Highest Sales & Profit
region_group= data.groupby(['category'])['profit'].mean()
region_group
```

```
category
Furniture           8.699327
Office Supplies    20.327050
Technology         78.752002
Name: profit, dtype: float64
```

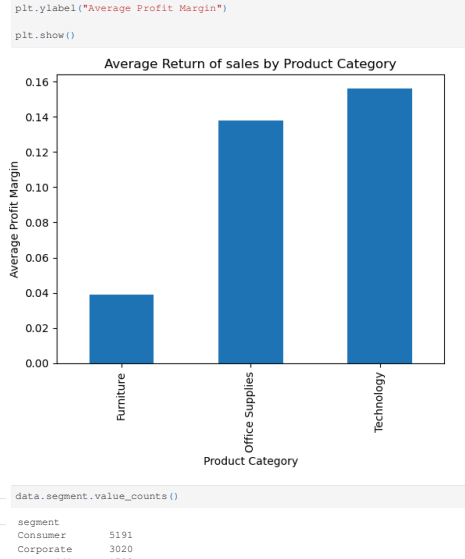
```
data['ROS']=data['profit'] / data['sales']
data
```

	row id	order id	order date	ship date	ship mode	customer id	customer name	segment	country	city	...	region	product id	category	sub-category	product name	sales
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.9600
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.6200
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.3680
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
9989	9990	CA-2014-110422	1/21/2014	1/23/2014	Second Class	TB-21400	Tom Boeckenhauer	Consumer	United States	Miami	...	South	FUR-FU-10001889	Furniture	Furnishings	Ultra Door Pull Handle	25.2480
9990	9991	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	Dave Brooks	Consumer	United States	Costa Mesa	...	West	FUR-FU-10000747	Furniture	Furnishings	Tenex B1-RE Series Chair Mats for Low Pile Car...	91.9600
9991	9992	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	Dave Brooks	Consumer	United States	Costa Mesa	...	West	TEC-PH-10003645	Technology	Phones	Aastra 57i VoIP phone	258.5760
9992	9993	CA-2017-121258	2/26/2017	3/3/2017	Standard Class	DB-13060	Dave Brooks	Consumer	United States	Costa Mesa	...	West	OFF-PA-10004041	Office Supplies	Paper	It's Hot Message Books with Stickers, 2 3/4" x 5"	29.6000
9993	9994	CA-2017-119914	5/4/2017	5/9/2017	Second Class	CC-12220	Chris Cortes	Consumer	United States	Westminster	...	West	OFF-AP-10002684	Office Supplies	Appliances	Acco 7-Outlet Masterpiece Power Center, Wihtou...	243.1600

9994 rows × 22 columns

```
data['ROS%']=round(data.ROS*100,2)
```

```
ros_group= data.groupby('category')['ROS'].mean()
ros_group.plot(kind='bar')

plt.title("Average Return of sales by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Average Profit Margin")

plt.show()
```



```
data.segment.value_counts()
```

```
segment
Consumer       5191
Corporate      3020
Home Office    1783
Name: count, dtype: int64
```

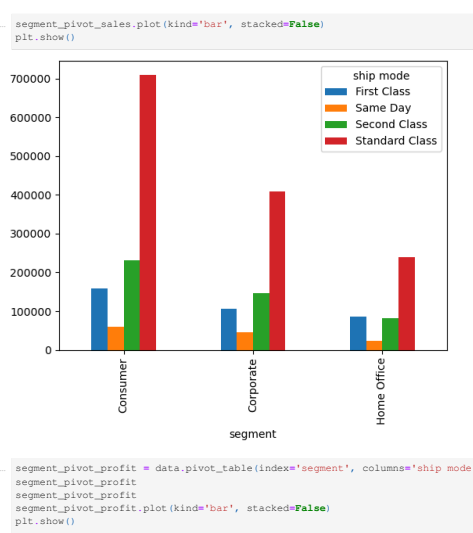
```
data['ship mode'].value_counts()
```

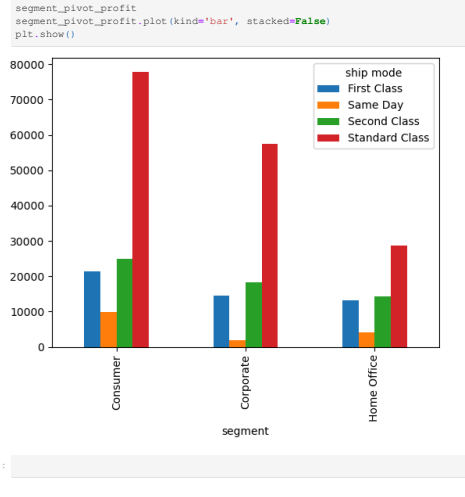
```
ship mode
Standard Class    5968
Second Class      1945
First Class       1538
Same Day           543
Name: count, dtype: int64
```

```
segment_pivot_sales = data.pivot_table(index='segment', columns='ship mode', values='sales', aggfunc='sum')
segment_pivot_sales
```

segment	First Class	Same Day	Second Class	Standard Class
Consumer	159168.9650	60596.359	231498.9496	710137.0714
Corporate	105858.4699	45121.323	146126.0388	409040.5351
Home Office	86400.9880	22645.443	81568.5810	239038.1365

```
segment_pivot_sales.plot(kind='bar', stacked=False)
plt.show()
```



```
segment_pivot_profit = data.pivot_table(index='segment', columns='ship mode', values='profit', aggfunc='sum')
segment_pivot_profit
segment_pivot_profit
segment_pivot_profit.plot(kind="bar", stacked=False)
plt.show()
```